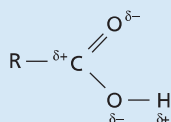
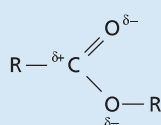


Carboxylic acids, esters and organic synthesis

Functional groups:



carboxylic acids



esters

In this topic we look at the major class of organic acids, the carboxylic acids, and at some of the compounds derived from them. Apart from acid-base reactions, their chemistry is dominated by nucleophilic substitution reactions, owing to the $\delta+$ carbon atom in the carbonyl group.

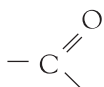
Learning outcomes

By the end of this topic, you will be able to:

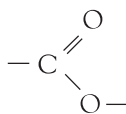
- 16.1.1** recall the reactions (reagents and conditions) by which alcohols can be produced:
- f hydrolysis of an ester using dilute acid or dilute alkali and heat
- 18.1.1** recall the reactions by which carboxylic acids can be produced:
- a the oxidation of primary alcohols and aldehydes with acidified $\text{K}_2\text{Cr}_2\text{O}_7$ or acidified KMnO_4 and refluxing
- b the hydrolysis of nitriles with dilute acid or dilute alkali followed by acidification
- c hydrolysis of esters with dilute acid or dilute alkali and heat followed by acidification
- 18.1.2** describe:
- a the redox reaction with reactive metals to produce a salt and $\text{H}_2(\text{g})$
- b the neutralisation reaction with alkalis to produce a salt and $\text{H}_2\text{O}(\text{l})$
- c the acid-base reaction with carbonates to produce a salt and $\text{H}_2\text{O}(\text{l})$ and $\text{CO}_2(\text{g})$
- d esterification with alcohols with concentrated H_2SO_4 as a catalyst
- e reduction by LiAlH_4 to form a primary alcohol
- 18.2.1** recall the reaction (reagents and conditions) by which esters can be produced:
- a the condensation reaction between an alcohol and a carboxylic acid with concentrated H_2SO_4 as a catalyst
- 18.2.2** describe the hydrolysis of esters by dilute acid and by dilute alkali and heat
- 19.2.3** describe the hydrolysis of nitriles with dilute acid or dilute alkali followed by acidification to produce a carboxylic acid
- 21.1.1** for an organic molecule containing several functional groups:
- a identify organic functional groups using the reactions in the syllabus
- b predict properties and reactions
- 21.1.2** devise multi-step synthetic routes for preparing organic molecules using the reactions in the syllabus
- 21.1.3** analyse a given synthetic route in terms of type of reaction and reagents used for each step of it, and possible by-products

19.1 Introduction

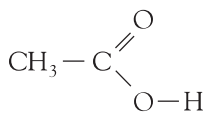
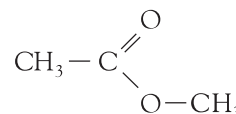
When the carbonyl group is directly joined to an oxygen atom, the **carboxyl group** is formed. This occurs in **carboxylic acids** and **esters**.



carbonyl group

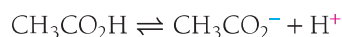


carboxyl group

carboxylic acid:
(ethanoic acid)ester:
(methyl ethanoate)

The reactions of the carbonyl group are drastically changed by the presence of the electronegative oxygen atom. These compounds have virtually none of the reactions of carbonyl compounds as described in Topic 18.

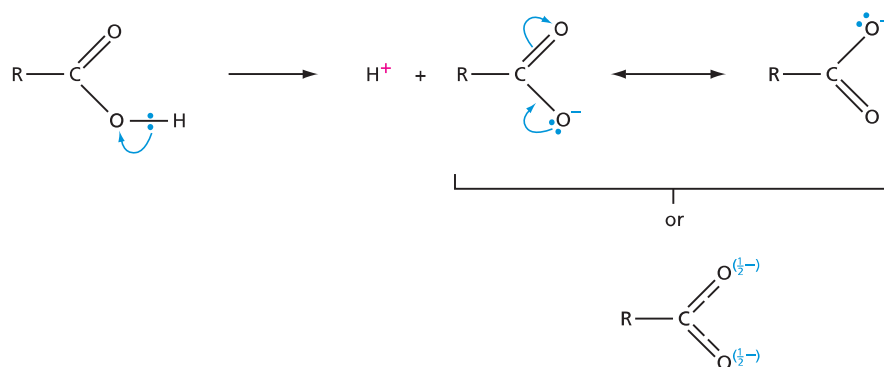
The reactivity of carboxylic acids is dominated by the tendency of the O—H bond to ionise to give hydrogen ions, hence the incorporation of the word 'acid' in their name. The extent of ionisation is small, however. For example, in a 1.0 mol dm⁻³ solution of ethanoic acid, about one molecule in 1000 (0.1%) is ionised:



Carboxylic acids are therefore classed as **weak acids** (see Section 6.4).

The O—H bond breaks heterolytically far more easily in carboxylic acids than in alcohols. The negative charge formed by heterolysis can be delocalised over two electronegative oxygen atoms in the carboxylate ion. This spreading out of charge invariably leads to a stabilisation of the anion (see Figure 19.1). The effect that changes in the structure of a carboxylic acid have on the stability of its anion, and hence its acidity, is dealt with in Section 28.1.

► Figure 19.1 The charge is spread in the carboxylate anion, stabilising the ion.



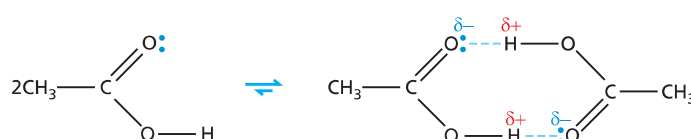
The carboxylic acid group is a strongly hydrogen-bonded one, both to carboxylic acid groups on other carboxylic acid molecules and to solvent molecules such as water. This has effects on two important properties: boiling points and solubilities. Table 19.1 compares these properties for carboxylic acids and the corresponding alcohols.

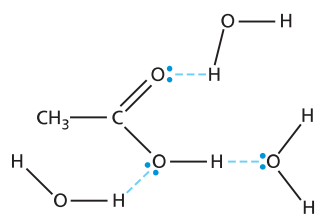
Number of carbon atoms	Alcohol			Carboxylic acid		
	Formula	Boiling point/°C	Solubility in 100 g of water/g	Formula	Boiling point/°C	Solubility in 100 g of water/g
1	CH ₃ OH	65	∞	HCO ₂ H	101	∞
2	CH ₃ CH ₂ OH	78	∞	CH ₃ CO ₂ H	118	∞
3	CH ₃ CH ₂ CH ₂ OH	97	∞	CH ₃ CH ₂ CO ₂ H	141	∞
4	CH ₃ CH ₂ CH ₂ CH ₂ OH	118	7.9	CH ₃ CH ₂ CH ₂ CO ₂ H	164	∞
5	CH ₃ CH ₂ CH ₂ CH ₂ CH ₂ OH	138	2.3	CH ₃ CH ₂ CH ₂ CH ₂ CO ₂ H	187	3.7

▲ Table 19.1 Hydrogen bonding increases the boiling points and solubilities in carboxylic acids.

Both in the pure liquid state, and in solution in non-hydrogen-bonding hydrocarbon solvents, carboxylic acids can form hydrogen-bonded dimers (see Figure 19.2).

► Figure 19.2 Carboxylic acids dimerise as two molecules of hydrogen bond to each other.



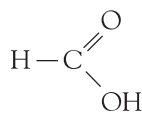


▲ Figure 19.3 Carboxylic acids hydrogen bond to water molecules.

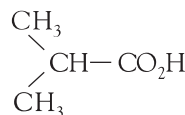
In water, both oxygen atoms and the —OH hydrogen atom of carboxylic acids can hydrogen bond with the solvent (see Figure 19.3).

19.2 Isomerism and nomenclature

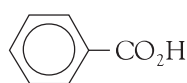
Carboxylic acids are named by finding the longest carbon chain that contains the acid functional group and adding the suffix ‘-oic acid’ to the stem. Side groups off the chain are named in the usual way:



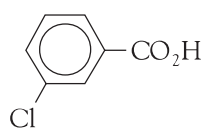
methanoic acid



2-methylpropanoic acid

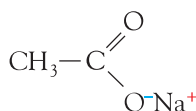


benzoic acid

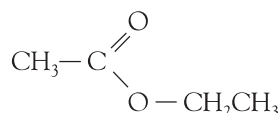


3-chlorobenzoic acid

Esters are named as alkyl derivatives of acids, similar to the naming of salts:



sodium ethanoate



ethyl ethanoate

The isomerism shown by acids is not usually associated with the functional group, but occurs in the carbon chain to which it is attached, as in the following two isomers with the formula $\text{C}_4\text{H}_8\text{O}_2$:

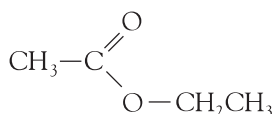


butanoic acid

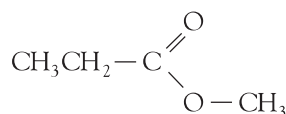


2-methylpropanoic acid

Esters, however, can show a particular form of isomerism, depending on the number of carbon atoms they have in the acid or alcohol part of the molecule. Two isomers of $\text{C}_4\text{H}_8\text{O}_2$ are shown below. Note that although the acid part is usually written first in the formula, by convention, it appears last in the name:



ethyl ethanoate



methyl propanoate

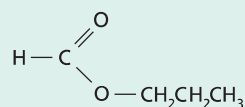
Question 19.1

Draw and name possible ester isomers with the formula $\text{C}_5\text{H}_{10}\text{O}_2$. How many ester isomers are there with this formula? Indicate in your answer which of the esters are chiral.

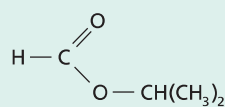
WORKED EXAMPLE 19A

Draw the structures of the other two isomeric esters with the formula $\text{C}_4\text{H}_8\text{O}_2$ and name them.

Answer



prop-1-yl methanoate



prop-2-yl methanoate



19.3 Reactions of carboxylic acids

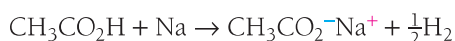
19

Reactions of the O—H group

The acidity of carboxylic acids has already been mentioned. They show most of the typical reactions of acids.

With metals

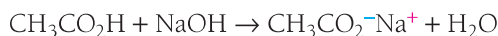
Ethanoic acid reacts with sodium metal in a redox reaction, liberating hydrogen gas (compare with alcohols, Section 17.3):



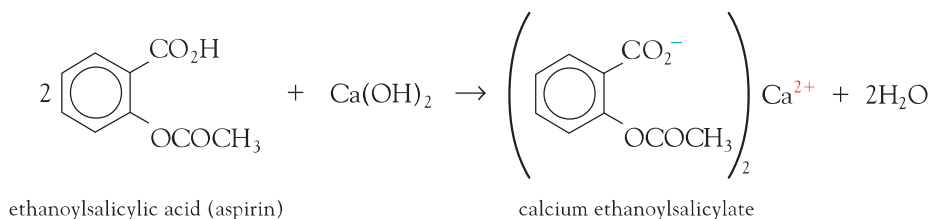
They also react in a similar way with other reactive metals, such as calcium and magnesium.

With sodium hydroxide

Unlike alcohols, carboxylic acids form salts with alkalis:



Salts are often much more soluble in water than the carboxylic acids from which they are derived. For example, soluble aspirin tablets contain the calcium salt of ethanoylsalicylic acid (see Section 27.6):



With sodium carbonate and sodium hydrogen carbonate

Like the usual inorganic acids, but unlike alcohols, carboxylic acids react with carbonates, liberating carbon dioxide gas:



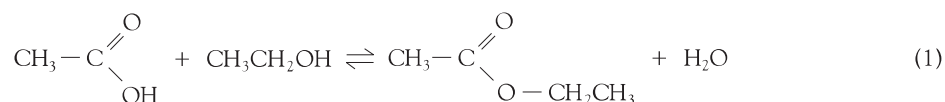
A similar reaction occurs with sodium hydrogen carbonate:



Reactions of the C—OH group

With alcohols in the presence of anhydrous acids

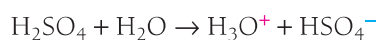
Carboxylic acids react with alcohols in an acid-catalysed equilibrium reaction to give esters:



The formation of an ester is an example of a **condensation reaction**. This is when two molecules join together with the elimination of a small molecule (often water). Typical reaction conditions are:

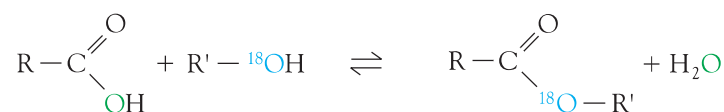
- » heat a mixture of carboxylic acid and alcohol with 0.1 mol equivalent of concentrated sulfuric acid under reflux for 4 hours, or
- » heat a mixture of carboxylic acid and alcohol under reflux for 2 hours, while passing dry hydrogen chloride gas through the mixture.

The reaction needs an acid catalyst and, like any equilibrium reaction, it can be made to go in either direction depending on the conditions. To encourage esterification, an excess of one of the reagents (often the alcohol) is used, and the water is removed as it is formed (often by the use of concentrated sulfuric acid):



This drives equilibrium (1) over to the right (see Section 9.3). To encourage the reverse reaction (the hydrolysis of the ester), the ester is heated under reflux with a dilute solution of sulfuric acid. This provides an excess of water to drive equilibrium (1) over to the left.

If an ester is made from an alcohol whose —OH group is labelled with an oxygen-18 atom, it is found that the ^{18}O is 100% incorporated into the ester. None of it appears in the water that is also produced. Therefore the water must come from combining the —OH from the carboxylic acid and the —H from the alcohol, rather than vice versa. That is,

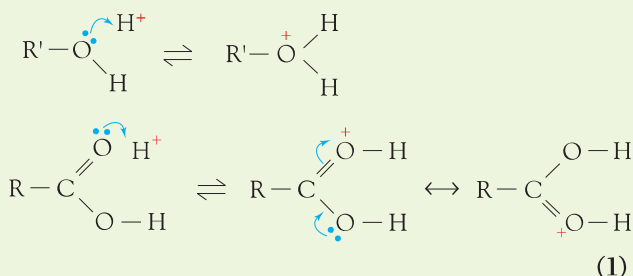


The following mechanism explains why this is the case.

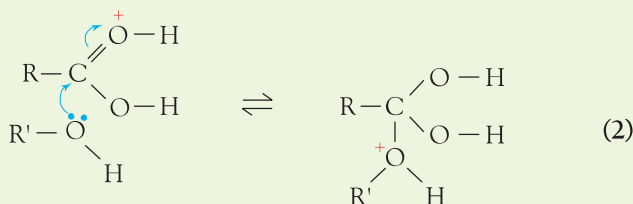
EXTENSION

The mechanism of esterification/hydrolysis

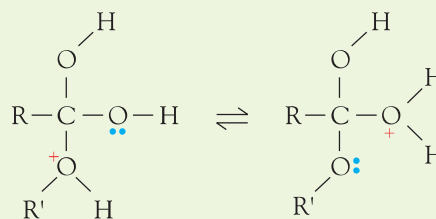
In a strongly acidic medium, both the alcohol and the carboxylic acid can undergo reversible protonation. However, only the protonated carboxylic acid undergoes further reaction.



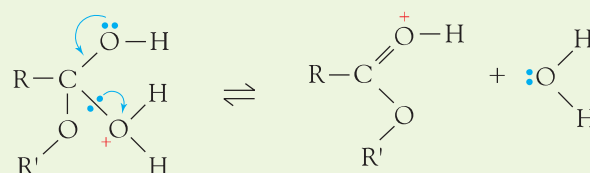
The protonation of the carboxylic acid in step (1) produces a highly δ^+ carbon atom which can undergo nucleophilic attack by a lone pair of electrons on the oxygen atom of an alcohol molecule:



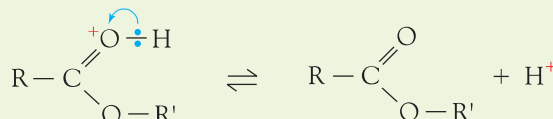
The three oxygen atoms in this intermediate cation are very similar in basicity. Any one of them could be attached to the proton, and proton transfer from one to another can occur readily:



This new cation can undergo a carbon–oxygen bond breaking, which is essentially the reverse of reaction (2), to give a molecule of water, along with the protonated ester:



Lastly, the cation loses a proton to regenerate the H^+ catalyst and form the ester product:



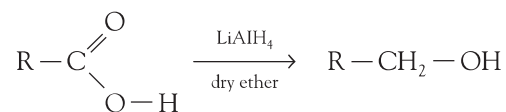
As mentioned above, the reaction is reversible. So is the mechanism. With an excess of water, esters undergo acid-catalysed hydrolysis. The steps in the mechanism for the hydrolysis reaction are the reverse of the above steps for esterification.

Question 19.2

Draw the mechanism for the acid-catalysed hydrolysis of methyl methanoate.

Reduction of the $-\text{CO}_2\text{H}$ group

Carboxylic acids can be reduced to alcohols by reacting with lithium tetrahydridoaluminate(III) (lithium aluminium hydride), LiAlH_4 , in dry ether.

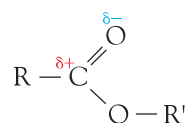


The reaction requires the powerful reducing agent LiAlH_4 ; neither NaBH_4 nor $\text{H}_2 + \text{Ni}$ are strong enough to reduce carboxylic acids.

19.4 Esters

Properties of esters

Esters have the functional group:



Many esters are liquids with sweet, 'fruity' smells. They are immiscible with, and usually less dense than, water. Despite containing two oxygen atoms, they do not form strong hydrogen bonds with water molecules. Neither do they form hydrogen bonds with other ester molecules (because they do not contain δ^+ hydrogen atoms). Their major intermolecular attraction is the instantaneous dipole-induced dipole (id-id) force, supplemented by a small dipole-dipole contribution. Their boiling points are therefore similar to, but a few degrees higher than, those of the alkanes of similar molecular mass. The position of the carboxyl group along the chain has little effect on the strength of intermolecular forces and hence boiling point (see Table 19.2).

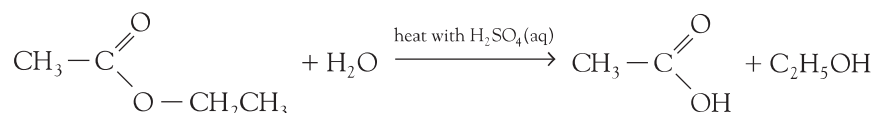
► Table 19.2 Esters have boiling points a little higher than those of corresponding alkanes.

Compound	M_r	Boiling point/ $^\circ\text{C}$
hexane	86	69
propyl methanoate	88	81
ethyl ethanoate	88	77
methyl propanoate	88	79

Reactions of esters

The most common type of reaction that esters undergo is nucleophilic substitution, illustrated here by their hydrolysis.

The hydrolysis of an ester is a slow process, taking several hours of heating under reflux with dilute aqueous acids:

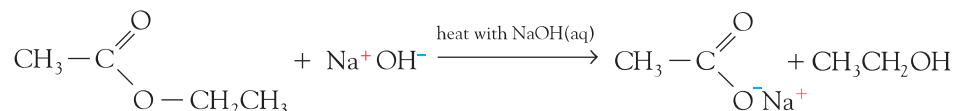


Question 19.3

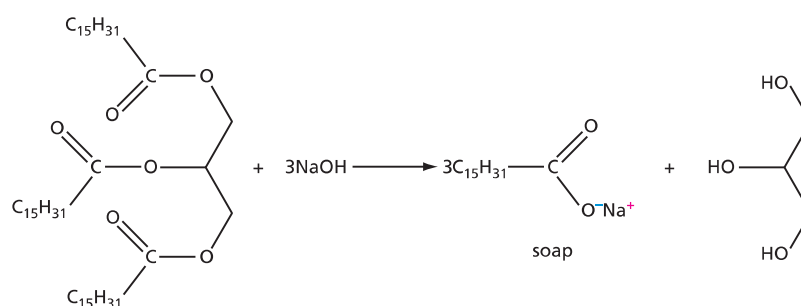
Suggest a mechanism for this reaction, using OH^- as the initial nucleophile, and $\text{C}_2\text{H}_5\text{O}^-$ as the base in the last step.

The mechanism of this acid-catalysed hydrolysis has been discussed in Section 19.3: it is the reverse of the acid-catalysed esterification described there.

Ester hydrolysis can also be carried out in alkaline solution. The reaction is quicker than in acid solution: OH^- is a stronger nucleophile than water. Additionally, it does not reach equilibrium, but goes to completion. This is because the carboxylic acid produced reacts with an excess of base to form the carboxylate anion:



The alkaline hydrolysis of esters has an important application in the manufacture of soap. Fats are glyceryl triesters of long-chain carboxylic acids. Heating fats with strong aqueous sodium hydroxide causes hydrolysis of the triesters. The sodium salts of the long-chain carboxylic acids are precipitated by adding salt (NaCl) to the mixture, and this solid is then washed and compressed into bars of soap (see Figure 19.4). Perfume and colour are also added.

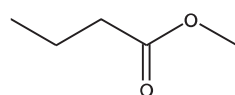


► Figure 19.4 Soap is made by the hydrolysis of the glyceryl triesters of fats.

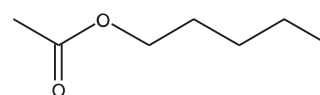
Uses of esters

Esters with small molecular masses find many uses as solvents for paints and varnishes. A number of natural perfumes, flavours and aromas are esters, and many of these are now manufactured commercially to flavour fruit drinks and sweets. It is the ethyl esters formed from alcohol and the various acids in wines that account for much of their 'nose', or aroma, in the glass.

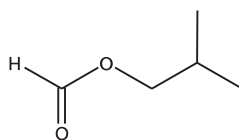
Some of the naturally occurring esters now made synthetically and used as artificial flavours in foodstuffs are shown in Figure 19.5.



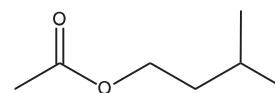
methyl butanoate (apple)



pentyl ethanoate (pear)



2-methylpropyl methanoate (raspberry)



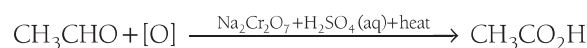
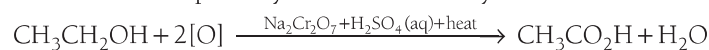
3-methylbutyl ethanoate (banana)

► Figure 19.5 Some esters used as flavouring agents

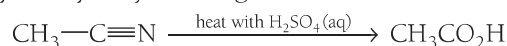
19.5 Preparing carboxylic acids

Carboxylic acids may be prepared by the following reactions.

► By the oxidation of primary alcohols or aldehydes:



- » By the hydrolysis of organic nitriles:



- » By the hydrolysis of esters. Although this reaction proceeds to give high yield, it is not normally a useful preparative reaction: esters are usually made from the carboxylic acids in the first place, so their hydrolysis forms no compounds of further use.



19.6 The synthesis of organic compounds



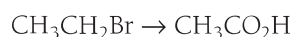
▲ Figure 19.6 A laboratory where new organic compounds are synthesised

Many organic compounds have important uses as pharmaceuticals, pesticides, perfumes and dyes. These compounds often have quite complicated structures, and most of them are manufactured by organic chemists from much simpler starting materials (see Figure 19.1). The science of organic synthesis is rather like building with molecular Lego™: a compound is constructed by combining a sequence of reactions – from two to perhaps 20 or so – which start from known, readily available chemicals and end up with the target molecule.

Organic synthesis requires much art and craft as well as science. If you have attempted an organic preparation in the laboratory you will know that the yield and purity of your product will often vary from those given in the ‘recipe’. Skill and practice are needed to perfect practical techniques. The reagents and conditions that give an excellent yield with one compound may not be as effective for another compound with an identical functional group.

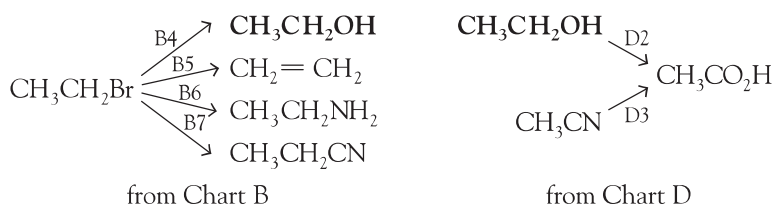
Nevertheless, it is possible to devise a multi-step method for synthesising a given organic compound by piecing together successive standard organic transformations. We shall illustrate this by making use of the reactions summarised in Charts A–E (see Section 19.9) and Table 19.3 (page 347–348).

For example, suppose we needed to devise a synthesis of ethanoic acid, starting from bromoethane:



The strategy is as follows:

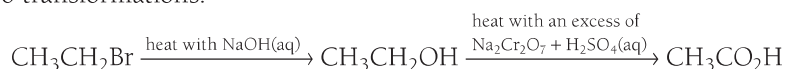
- 1 Use Chart B (page 348) to work out the structures of all the compounds that can be made from bromoethane in one step.
- 2 Next we use Chart D (page 348) to work out the structures of all the compounds that could be used to make ethanoic acid.
- 3 Then we see whether there is a compound that is common between both charts.



The compound **ethanol** is common to both, so a viable synthetic route would have ethanol as an intermediate:



Finally, from Table 19.3, we find the reagents and conditions required to carry out these two transformations:



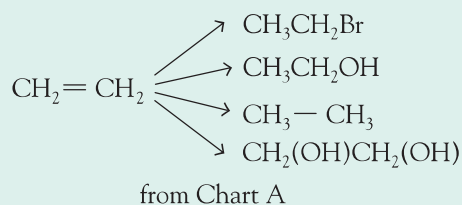
Steps 1 and 2 above using Charts A–E may need to be repeated if a common compound is not found.

WORKED EXAMPLE 19B

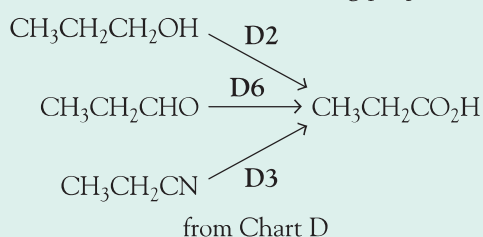
Devise a synthesis of propanoic acid, $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$, from ethene, C_2H_4 .

Answer

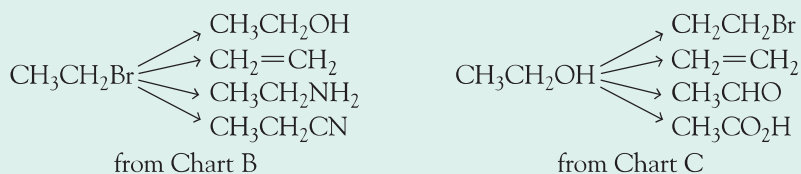
Chart A (page 347) shows that there are four reactions of ethene that may be useful:



Next, considering the target molecule, we can see from Chart D (page 350) that there are three methods of making propanoic acid:

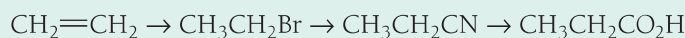


None of the three starting materials for making propanoic acid is the same as the four products from ethene, so the synthesis will need another step. Therefore, we take each of the four products derived from ethene, and use the charts to carry out the same analysis on those products. What compounds can we make from each one in turn?

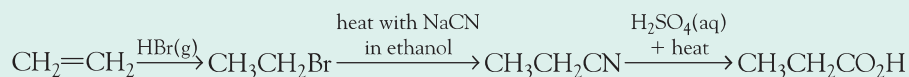


We stop immediately at Chart B, as we can see that there is a compound that is one of the three from which propanoic acid could be made: propanenitrile, $\text{CH}_3\text{CH}_2\text{CN}$.

The synthetic route therefore includes two intermediates:



Adding the reagents and conditions, the whole synthesis can be described as follows:



(Note that a 'shortcut' to the process here is given by the fact that the chain of the product contains one more carbon atom than the starting material: the synthesis must have included a step involving a nitrile.)

Questions 19.4

Devise syntheses for the following compounds, starting with the specified compounds:

- 1 $\text{CH}_3\text{CH}(\text{OH})\text{CN}$ from $\text{CH}_3\text{CH}_2\text{Br}$ (three steps)
- 2 $\text{CH}_3\text{CO}_2\text{CH}_3$ from CH_3CN and CH_3OH (two steps)
- 3 $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ from $\text{CH}_3\text{CH}_2\text{OH}$ (four steps)
- 4 $(\text{CH}_3)_2\text{CHCO}_2\text{CH}(\text{CH}_3)_2$ with most of its carbon atoms coming from propene, $\text{CH}_3\text{CH}=\text{CH}_2$ (five steps).



19.7 Analysing synthetic routes

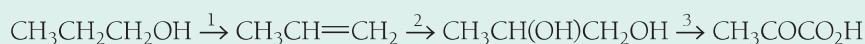
The table and charts in Section 19.9 can be also be used in reverse, to analyse a given synthetic route, identifying the types of reaction and the reagents used for each step.

Two examples will show how this can be done.

19

WORKED EXAMPLE 19C

Identify the types of reaction and the reagents for each step in the following synthesis of 2-oxopropanoic acid:



Answer

Step 1

The transformation of an alcohol to an alkene appears as reaction A2 on Chart A and also C5 on Chart C. Consulting Table 19.3 we see that this is an **elimination reaction**, and can be carried out either by heating the alcohol with concentrated sulfuric acid or by passing its vapour over hot Al_2O_3 .

Step 2

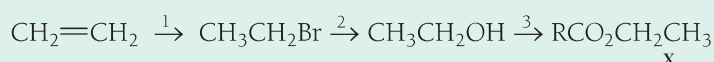
The transformation of an alkene into a diol only appears on Chart A, as reaction A6. Consulting Table 19.3 gives the reagent as $\text{KMnO}_4(\text{aq})$ at room temperature, and it is classified as an **oxidation reaction**.

Step 3

The production of carboxylic acid (and a ketone) from an alcohol appears on Chart C as reaction C6 or C7, and on Chart D as D1 or D2. Table 19.3 gives the reagent as $\text{Na}_2\text{Cr}_2\text{O}_7(\text{aq}) + \text{H}_2\text{SO}_4(\text{aq}) + \text{heat}$, and the reaction is an **oxidation reaction**.

WORKED EXAMPLE 19D

Identify the types of reaction and the reagents for each step in the following synthesis of the ethyl ester X.



Answer

Step 1

Conversion of an alkene into a bromoalkane appears in Chart A as reaction A3 and in Chart B as reaction B3. Table 19.3 shows this to be an **electrophilic addition** reaction, using $\text{HBr}(\text{g})$ as the reagent at room temperature.

Step 2

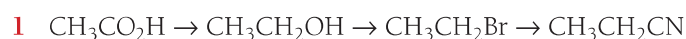
Bromoalkanes and alcohols appear in Chart B as reaction B4 and Chart C as reaction C1. Table 19.3 lists the conversion from one to the other as involving the reagent $\text{NaOH}(\text{aq}) + \text{heat}$, in a **nucleophilic substitution** reaction.

Step 3

The conversion of an alcohol into an ester appears as reaction C8 in Chart C (and as D7 in Chart D). From Table 19.3 we can see that the reagent and conditions would be heating with RCO_2H and conc. H_2SO_4 , and it is an **esterification** reaction.

Questions 19.5

Identify the types of reaction and the reagents for each step in the following synthetic transformations:





19.8 Multifunctional compounds

Many organic compounds contain several functional groups which may be the same as, or different from, each other. Such compounds will demonstrate the reactions of *all* their functional groups and if a given reagent reacts with a multifunctional compound, a reaction will usually occur with each and every functional group that potentially could react.

For example, the compound 1,3-dihydroxybutane, $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{OH}$, contains two alcohol —OH groups: one is a primary alcohol and the other is a secondary alcohol. On oxidation, the primary alcohol group could be converted into either an aldehyde or a carboxylic acid, depending on the reaction conditions (see Section 17.3). The secondary alcohol group will be oxidised to a ketone.



If a functional group can be converted into another functional group by two or more different reagents, it might be possible to select a reagent that reacts only with that group in the presence of another.

WORKED EXAMPLE 19E

Suggest a suitable reagent (one in each case) for converting 4-oxopent-1-ene into

- a 4-hydroxypent-1-ene
- b pentan-2-ol.

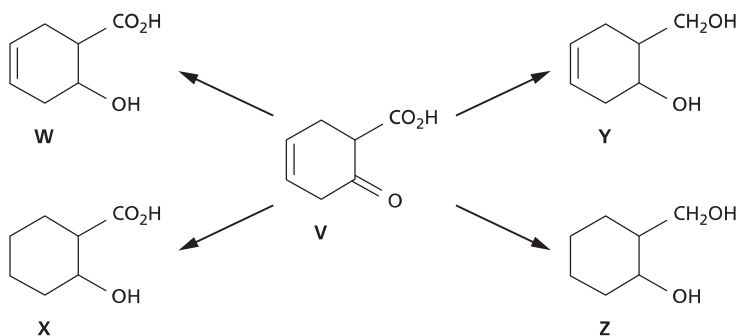
Answer

- a Hydrogen over a nickel catalyst will reduce a $\text{C}=\text{C}$ double bond to a C—C single bond, but will also reduce a carbonyl, $\text{C}=\text{O}$, to an alcohol, —CH(OH)— .
- b However, NaBH_4 will only reduce carbonyl compounds; it will not reduce isolated $\text{C}=\text{C}$ double bonds.



Question 19.6

Suggest suitable reagents for converting 2-oxo-cyclohex-4-enoic acid, **V**, into each of the four reduction products **W–Z**.



19.9 Summary of organic transformations

▼ Table 19.3 Reagents for organic reactions (AS)

Reagents for organic reactions

Reference (see key below)	Reagents and conditions	Reaction number on chart				
		Chart A	Chart B	Chart C	Chart D	Chart E
		Alkenes	Bromo- alkanes	Alcohols	Aldehydes and acids	Methylketones
EA1	HBr(g) at RT	A3	B3			
EA2	H ₂ SO ₄ (aq) at RT	A4		C2		E1
ET1	R'-OH + conc. H ₂ SO ₄ + heat				D7	
ET1	R'CO ₂ H + conc. H ₂ SO ₄ + heat			C8		
FR1	X ₂ (g) + light (X = Cl or Br)		B2			
HY1	H ₂ SO ₄ (aq) + heat				D3 D8 D10	
NA1	HCN + NaCN in ethanol/water				D5	E5
NS1	NaOH(aq) + heat		B4	C1		
NS2	NH ₃ in ethanol + heat under pressure		B6			
NS3	NaCN in ethanol + heat		B7			
NS4	HBr(conc.) or NaBr + conc. H ₂ SO ₄ + heat		B1	C4		
O1	Na ₂ Cr ₂ O ₇ (aq) + H ₂ SO ₄ (aq) + heat			C6 C7	D1 D2 D6	E2
O2	KMnO ₄ (aq) at RT	A6				
O3	KMnO ₄ (conc.) + H ₂ SO ₄ (aq) + heat	A7				
O4	I ₂ (aq) + OH ⁻ (aq) + warm					E4
R1	H ₂ (g) + Ni catalyst at RT	A5		C3	D4	E3
R2	NaBH ₄ in aqueous methanol + heat			C3	D4	E3
R3	LiAlH ₄ in dry ether			C3 C9	D4 D9	E3
X1	NaOH in ethanol + heat	A1	B5			
X2	Al ₂ O ₃ + heat	A2		C5		
X3	conc. H ₂ SO ₄ + heat	A2		C5		

Key:

EA	electrophilic addition
ET	esterification
FR	free radical
HY	hydrolysis
NA	nucleophilic addition
NS	nucleophilic substitution
O	oxidation
R	reduction
X	elimination
RT	room temperature

Charts for organic syntheses

Chart A: Synthetic routes involving alkenes

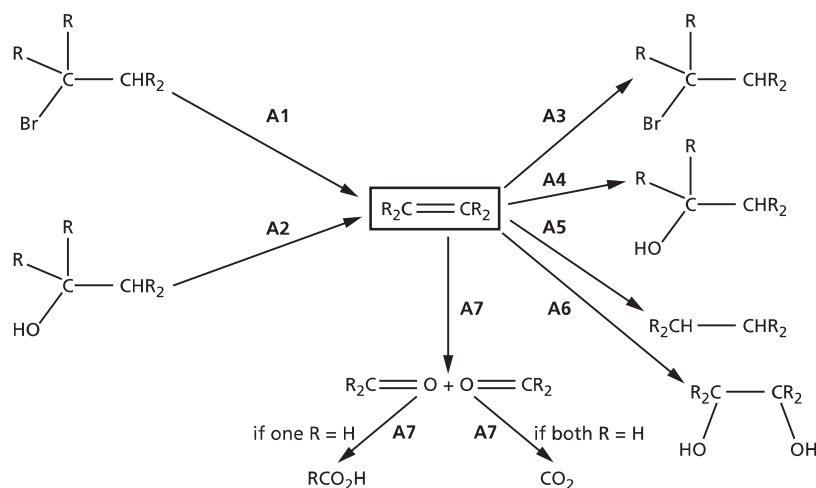
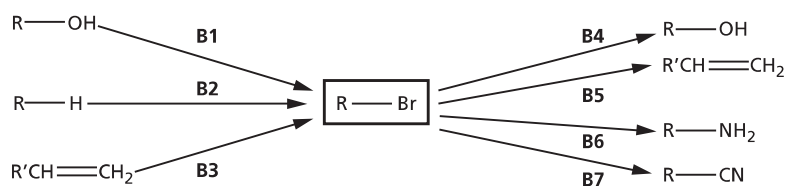
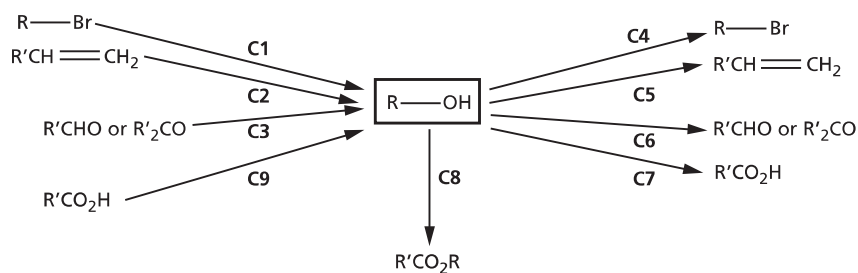
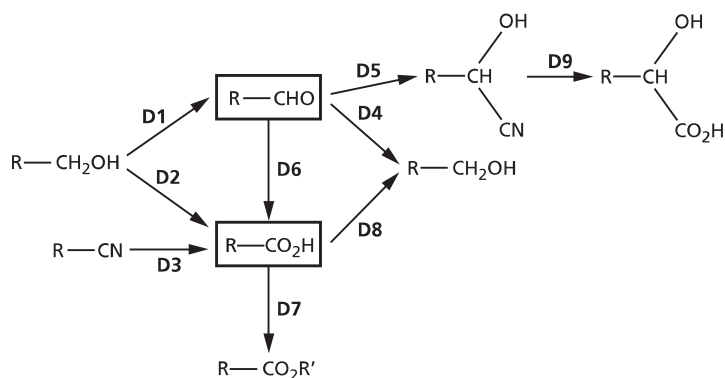
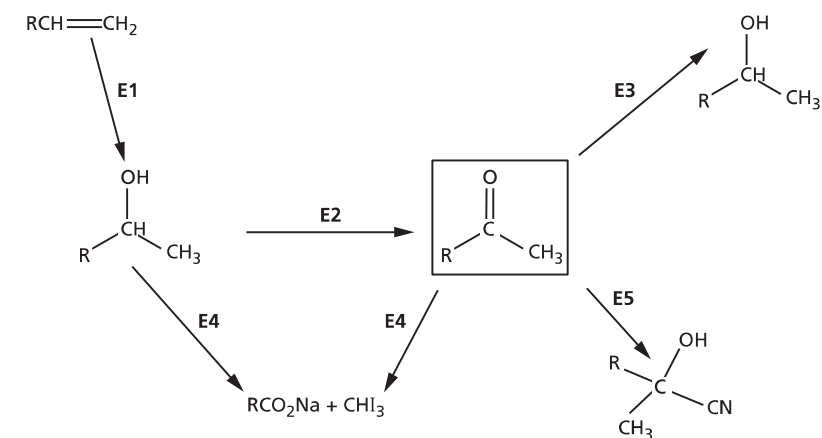


Chart B: Synthetic routes involving bromoalkanes (or chloroalkanes)*Chart C: Synthetic routes involving alcohols**Chart D: Synthetic routes involving aldehydes and acids**Chart E: Synthetic routes involving methyl ketones*

SUMMARY

- » The **carboxylic acids**, RCO_2H , are weak acids, ionising to the extent of 1% or less in water.
- » Carboxylic acids react with sodium metal giving off $\text{H}_2(\text{g})$ and carbonates to give $\text{CO}_2(\text{g})$.
- » Carboxylic acids react with alcohols to form **esters**.
- » Esters undergo acid- or base-catalysed hydrolysis.
- » The simple reactions of organic compounds can be 'bolted together' to allow the synthesis of more complex organic compounds.

Key reactions you should know

- » Carboxylic acids:

$$\text{RCO}_2\text{H} + \text{Na} \rightarrow \text{RCO}_2^-\text{Na}^+ + \frac{1}{2}\text{H}_2$$

$$\text{RCO}_2\text{H} + \text{NaOH} \rightarrow \text{RCO}_2^-\text{Na}^+ + \text{H}_2\text{O}$$

$$2\text{RCO}_2\text{H} + \text{Na}_2\text{CO}_3 \rightarrow 2\text{RCO}_2^-\text{Na}^+ + \text{CO}_2 + \text{H}_2\text{O}$$

$$\text{RCO}_2\text{H} + \text{R}'\text{OH} \xrightarrow{\text{heat with conc. H}^+} \text{RCO}_2\text{R}' + \text{H}_2\text{O}$$
- » Esters:

$$\text{RCO}_2\text{R}' \xrightarrow{\text{heat with H}_3\text{O}^+ \text{ or OH}^-(\text{aq})} \text{RCO}_2\text{H} + \text{R}'\text{OH}$$

END OF TOPIC QUESTIONS

Please see the Data Booklet on the website www.hoddereducation.co.uk/cambridgeextras for any data you may need to answer these questions.

- 1 Isomerism occurs in many organic compounds. The two main forms of isomerism are structural isomerism and stereoisomerism. Many organic compounds that occur naturally have molecules that can show stereoisomerism, that is *cis-trans* or optical isomerism.

- i Explain what is meant by *structural isomerism*.
- ii State two different features of molecules that can give rise to stereoisomerism.

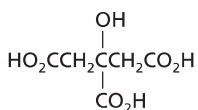
[3]

Unripe fruit often contains polycarboxylic acids, that is acids with more than one carboxylic acid group in their molecule. One of these acids is commonly known as tartaric acid, $\text{HO}_2\text{CCH}(\text{OH})\text{CH}(\text{OH})\text{CO}_2\text{H}$.

- b Give the structural formula of the organic compound produced when tartaric acid is reacted with an excess of NaHCO_3 .

[1]

Another acid present in unripe fruit is citric acid,



- c Does citric acid show optical isomerism? Explain your answer.

[1]

A third polycarboxylic acid present in unripe fruit is a colourless crystalline solid, **W**, which has the following composition by mass: C, 35.8%; H, 4.5%; O, 59.7%.

- d i Show by calculation that the empirical formula of **W** is $\text{C}_4\text{H}_6\text{O}_5$.
- ii The M_r of **W** is 134. Use this value to determine the molecular formula of **W**.

[3]

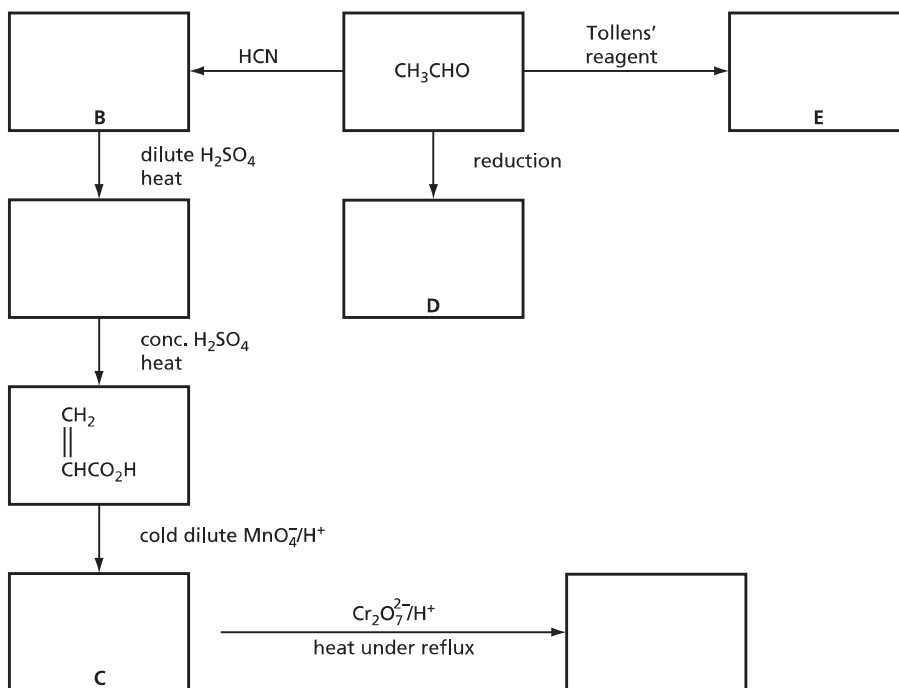
A sample of **W** of mass 1.97 g was dissolved in water and the resulting solution titrated with 1.00 mol dm^{-3} NaOH . 29.4 cm^3 were required for complete neutralisation.

- e i Use these data to deduce the number of carboxylic acid groups present in one molecule of **W**.
- ii Suggest the displayed formula of **W**.

[5]

[Cambridge International AS & A Level Chemistry 9701, Paper 21 Q5 June 2010]

- 2 a Draw and complete the following reaction scheme which starts with ethanal. In **each empty** box, write the **structural formula** of the organic compound that would be formed. [6]

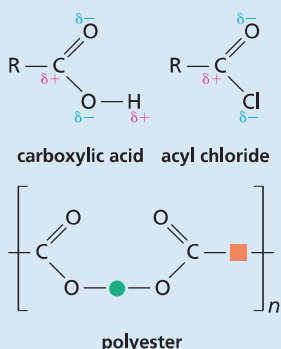


- b Write the structural formula for the organic compound formed when, under suitable conditions,
- compound **C** reacts with compound **D**
 - compound **C** reacts with compound **E**. [2]
- c Compound **B** is chiral. Draw displayed formulae of the two optical isomers of compound **B**, indicating with an asterisk (*) the chiral carbon atom. [3]

[Cambridge International AS & A Level Chemistry 9701, Paper 2 Q4 June 2009]

Carboxylic acids: derivatives and further reactions

Functional groups:



In this topic we look at the effect of structure on the acidity of carboxylic acids, the oxidation of two of them, and at further reactions of their important derivative, the acyl chlorides.

Learning outcomes

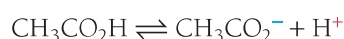
By the end of this topic, you will be able to:

- 29.2.1** understand and use the following terminology associated with types of organic mechanisms:
- b** addition–elimination
- 33.1.2** describe the reaction of carboxylic acids with PCl_3 and heat, PCl_5 , or SOCl_2 to form acyl chlorides
- 33.1.3** recognise that some carboxylic acids can be further oxidised:
- a** the oxidation of methanoic acid, HCOOH , with Fehling's reagent or Tollens' reagent or acidified KMnO_4 or acidified $\text{K}_2\text{Cr}_2\text{O}_7$ to carbon dioxide and water
- b** the oxidation of ethanedioic acid, HOOC-COOH , with warm, acidified KMnO_4 to carbon dioxide
- 33.1.4** describe and explain the relative acidities of carboxylic acids, phenols and alcohols
- 33.1.5** describe and explain the relative acidities of chlorine-substituted carboxylic acids
- 33.2.1** recall the reaction by which esters can be produced:
- a** reaction of alcohols with acyl chlorides, using the formation of ethyl ethanoate and phenyl benzoate as examples
- 33.3.1** recall the reactions (reagents and conditions) by which acyl chlorides can be prepared:
- a** reaction of carboxylic acids with PCl_3 and heat, PCl_5 , or SOCl_2
- 33.3.2** describe the following reactions of acyl chlorides:
- a** hydrolysis on addition of water at room temperature to give the carboxylic acid and HCl
- b** reaction with an alcohol at room temperature to produce an ester and HCl
- c** reaction with phenol at room temperature to produce an ester and HCl
- d** reaction with ammonia at room temperature to produce an amide and HCl
- e** reaction with a primary or secondary amine at room temperature to produce an amide and HCl
- 33.3.3** describe the addition–elimination mechanism of acyl chlorides
- 33.3.4** explain the relative ease of hydrolysis of acyl chlorides, alkyl chlorides and halogenoarenes (aryl chlorides)



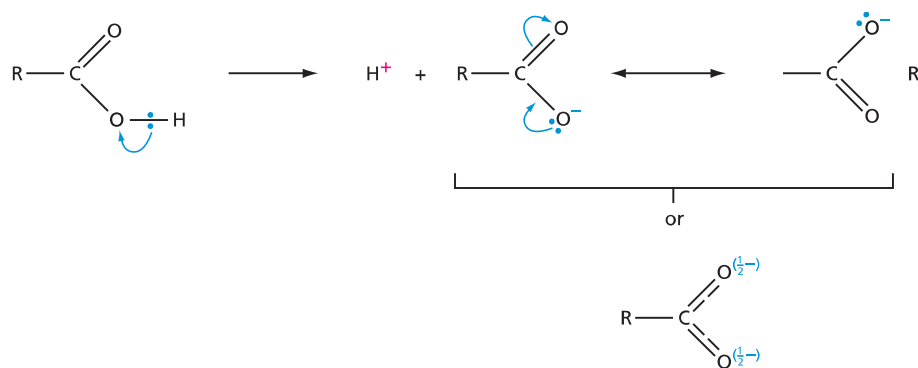
28.1 The effect of structure on acid strength

As we saw in Topic 19, carboxylic acids are weak acids, and dissociate to a small extent when in aqueous solution:



They are more acidic than alcohols because the negative charge on the anion can be delocalised over two electronegative oxygen atoms. (see Figure 28.1 and Figure 27.6).

- **Figure 28.1** In the carboxylate anion, the charge is spread, increasing the stability of the ion.



The increasing acidities of alcohols, phenols and carboxylic acids are explained by the increasing ability of the molecular structures to delocalise the negative charge in the alkoxide, phenoxide and carboxylate anions. Their relative acidities are demonstrated by their different reactions with sodium, sodium hydroxide and sodium carbonate (see Table 28.1).

- **Table 28.1** The use of Na, NaOH and Na₂CO₃ to illustrate the relative acidities of alcohols, phenol and carboxylic acids.

Reagent	Observation with		
	Hexanol	Phenol	Hexanoic acid
Na(s)	H ₂ (g) evolved	H ₂ (g) evolved	H ₂ (g) evolved
NaOH(aq)	no reaction	dissolves	dissolves
Na ₂ CO ₃ (aq)	no reaction	no reaction	CO ₂ (g) evolved

Atoms or groups that draw electrons away from the —CO₂[−] group will help the anion to form, and this causes the acid to be more dissociated (that is, to become a stronger acid). On the other hand, groups that donate electrons to the —CO₂[−] group will cause the acid to become weaker (see Table 28.2).

Electron-donating groups decrease the acid strength of carboxylic acids, whereas electron-withdrawing groups increase their acid strength.

- **Table 28.2** The acidity of some carboxylic acids. Remember from Topic 24 that $pK_a = -\log_{10} K_a$, where

$$K_a = \frac{[H^+(aq)][A^-(aq)]}{[HA(aq)]}$$

for acids undergoing the dissociation $HA \rightleftharpoons H^+ + A^-$

Formula of acid	pK _a	Percentage dissociation in 1.0 mol dm ^{−3} aqueous solution
	3.75	1.3%
	4.76	0.42%
	4.87	0.36%
	2.87	3.7%
	1.26	21%
	0.66	59%

WORKED EXAMPLE 28A

The pK_a of fluoroethanoic acid is 2.57.

- a** Does this mean it is a stronger or a weaker acid than chloroethanoic acid? Explain your answer.
b What pK_a would you expect for difluoroethanoic acid?

Answer

- a** The lower the pK_a , the larger is K_a . This means that the acid is more dissociated and therefore stronger. Hence fluoroethanoic acid is a stronger acid than chloroethanoic acid. This is due to the greater electron-withdrawing ability of the highly electronegative fluorine atom.
b Difluoroethanoic acid would be expected to have a lower pK_a than dichloroethanoic acid: about 1.0.

Questions 28.1

- 1** Use the data in Table 28.2 to predict the pK_a values of:
a $(CH_3)_3C-CO_2H$
b $Cl-CH_2-CH_2-CO_2H$.
 Explain your reasoning.
2 Explain why the pK_a values of 2-chlorobutanoic acid ($pK_a = 2.86$) and 4-chlorobutanoic acid ($pK_a = 4.53$) differ so much. Predict the pK_a value for 3-chlorobutanoic acid.

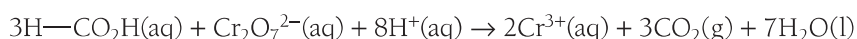
The effect of electron-donating or electron-withdrawing groups is seen even when such groups are situated on the opposite side of a benzene ring to the $-CO_2H$ group (see Table 28.3).

► **Table 28.3** The effect on pK_a for benzoic acid of electron-donating and electron-withdrawing groups

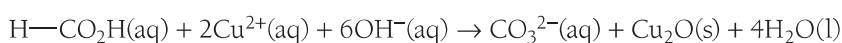
Formula of acid	pK_a	Percentage dissociation in 1.0 mol dm^{-3} aqueous solution
	4.20	0.80%
	4.37	0.73%
	3.99	1.0%

28.2 The further oxidation of some carboxylic acids

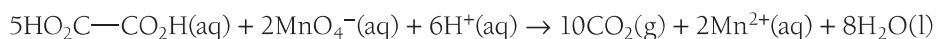
Unlike other aliphatic carboxylic acids, **methanoic acid** does not contain an alkyl chain attached to the $-CO_2H$ group. Instead, the $H-CO$ group, consisting of a hydrogen atom attached to a carbonyl group, has some reactions in common with aldehydes. In particular, it undergoes oxidation with common oxidants such as acidified dichromate(VI).



It also reacts with both Fehling's solution and Tollens' reagent (see page 331).



Ethanedioic acid contains two adjacent carbonyl groups. The proximity of two δ^+ carbon atoms weakens the C-C bond sufficiently for the molecule to be readily oxidised by warm acidified manganate(VII).



This forms the basis of an accurate volumetric method of analysing solutions containing ethanedioic acid or its salts.

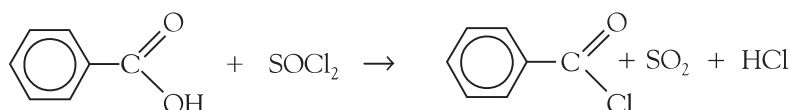
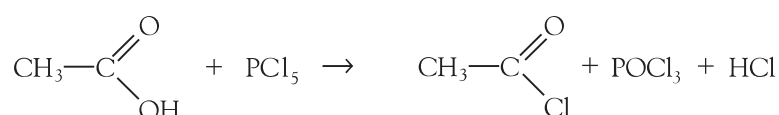
In a similar type of reaction, **α -ketoacids** are oxidised by alkaline H_2O_2 with the loss of a carbon atom.



28.3 Acyl chlorides

With phosphorus(V) chloride or sulfur dichloride oxide

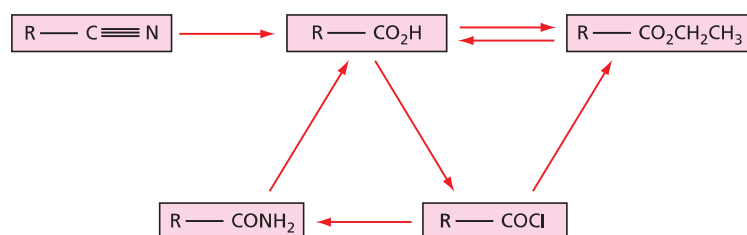
Carboxylic acids react with phosphorus(V) chloride or sulfur dichloride oxide in exactly the same way as do alcohols (see Section 17.3). Acyl chlorides are produced:



Many acid derivatives can be prepared from acyl chlorides. Figure 28.2 shows a chart summarising the interrelationships between the various derivatives.

Question 28.2

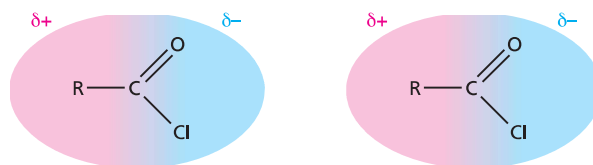
Copy the chart in Figure 28.2 and write on each arrow the correct reagents and conditions for each reaction.



▲ Figure 28.2 Interconversions between carboxylic acids and their derivatives

Properties of acyl chlorides

Acyl chlorides are usually liquids which fume in moist air. They are immiscible with water, but react slowly with it and eventually dissolve (see below). They are not hydrogen bonded: their main intermolecular attractions are a combination of van der Waals' forces and dipole-dipole forces (see Figure 28.3).



▲ Figure 28.3 Dipole-dipole attractions in acyl chlorides

The extra electron-withdrawing effect of the carbonyl group, resulting in extra dipole-dipole attractions, has the effect of increasing the boiling point by about 10–15 °C compared with the boiling point of the halogenoalkane with similar shape (see Table 28.4).

► **Table 28.4** Strong dipole–dipole attractions increase the boiling points in acyl chlorides.

Chloroalkane	M_r	Boiling point/ $^{\circ}\text{C}$	Acyl chloride	M_r	Boiling point/ $^{\circ}\text{C}$	Difference / $^{\circ}\text{C}$
	78.5	37		78.5	52	15
	92.5	68		92.5	80	12

Question 28.3

Gather together all the information given in Sections 16.3, 27.4 and this section to write an explanation of why the rates of hydrolysis of these three chloro compounds differ in this way. Would you expect the same relative rates with NH_3 or NaOH as the reagent, instead of H_2O ?

Reactions of acyl chlorides

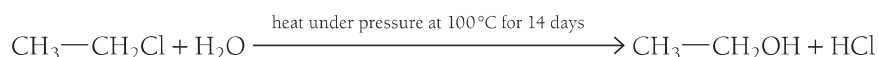
The electronegativity of the oxygen, and the easily polarised $\text{C}=\text{O}$ double bond, have a dramatic effect on the reactivity of acyl chlorides compared with that of chloroalkanes.

With water

Acyl chlorides react readily with water:

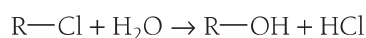


Compare this with the conditions for the hydrolysis of a halogenoalkane:

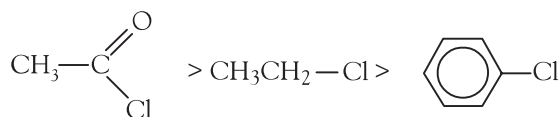


As a result of the speedy reaction with water acyl chlorides produce steamy fumes of $\text{HCl}(\text{g})$ when a few drops of water are added to them. This is a good test for them.

Together with what we have already seen in Sections 16.3 and 27.4, we note that in the reaction:

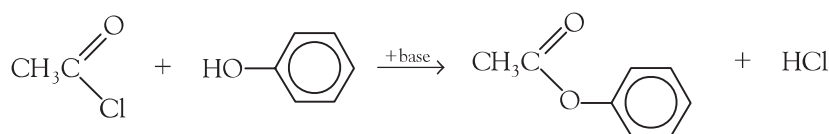
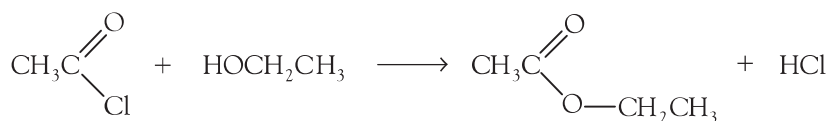


the relative rates of hydrolysis are as follows:

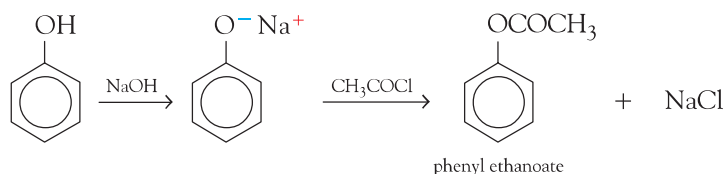


With alcohols or phenols

Acyl chlorides react readily with alcohols or phenols, forming esters:



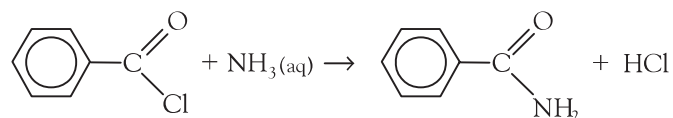
As mentioned in Section 27.5, phenols are not so nucleophilic as alcohols, because the lone pair on the oxygen atom is delocalised over the ring. The acylation of phenols is therefore usually carried out under basic conditions (NaOH or pyridine have been used), when the more nucleophilic phenoxide ion is formed as an intermediate:



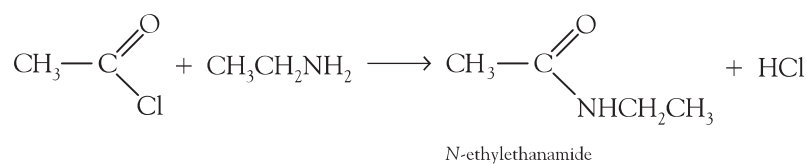
Because phenols do not react directly with carboxylic acids, this is the only method for preparing phenyl esters (see Topic 27, page 510).

With ammonia and amines

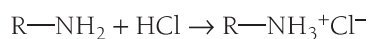
Acyl chlorides react with ammonia, forming amides. The reaction is vigorous, and white smoky fumes of $\text{NH}_4\text{Cl(s)}$ are often seen if an excess of ammonia is used.



They react with primary amines (see Topic 29), forming substituted amides:



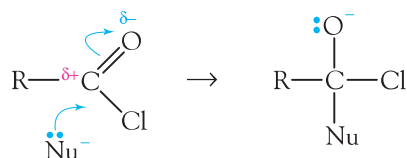
The hydrogen chloride produced reacts with another molecule of amine or ammonia in an acid–base reaction:



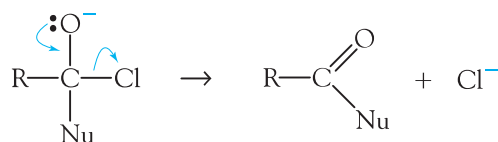
Therefore an excess of amine or ammonia is used to ensure complete reaction. The reactions of amides are described in Section 29.5.

The addition–elimination mechanism of the reactions of acyl chlorides

The carbonyl group in acyl chlorides can undergo nucleophilic addition in a similar manner to carbonyl compounds:

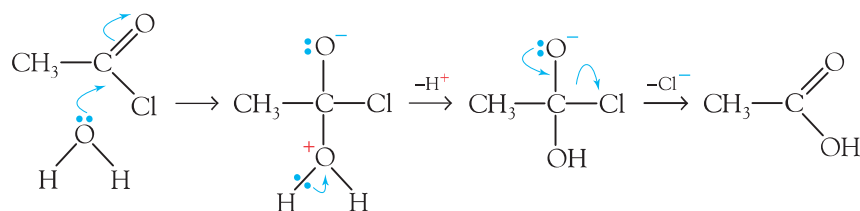


Unlike carbonyl compounds, however, acyl chlorides are provided with an easily removed leaving group, the chloride ion:



Nucleophilic substitution has taken place, by a mechanism involving addition, followed by elimination.

If the nucleophile is water, the carboxylic acid is formed:



Question 28.4

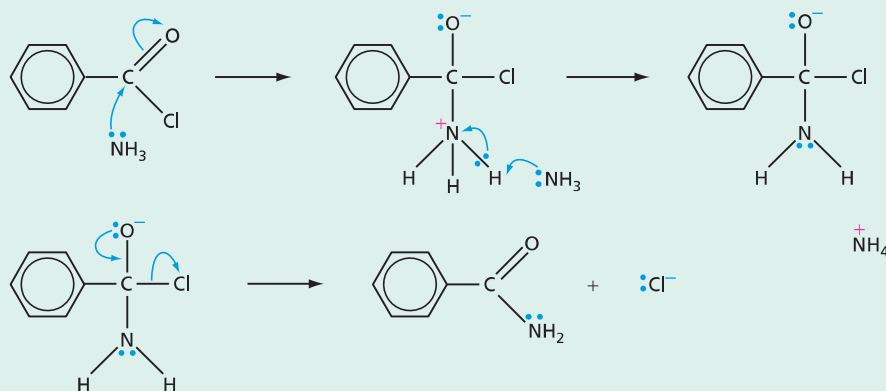
Predict the products of, and suggest a mechanism for, the reaction between propan-1-ol and propanoyl chloride.

WORKED EXAMPLE 28B

Draw the mechanism for the reaction between ammonia and benzoyl chloride:



Answer



Explaining the relative reactivity of acyl chlorides, chloroalkanes and chloroarenes

As we saw in Section 27.4, chlorobenzene does not undergo hydrolysis with water, or even with concentrated sodium hydroxide solution, due to the overlap of the lone pair on chlorine making the C—Cl bond stronger. Acyl chlorides, on the other hand, react much more readily than chloroalkanes with water or hydroxide. There are two reasons for this.

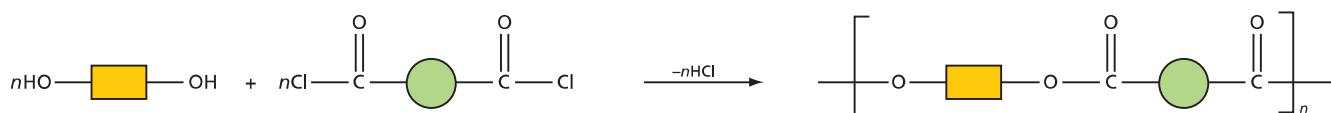
- 1 As we have just seen, the unsaturated nature of the C=O bond allows the C—OH bond to be formed before the C—Cl bond has broken. This **addition–elimination** mechanism therefore has a lower activation energy than the straight $\text{S}_{\text{N}}2$ reaction.
- 2 The carbon atom in acyl chlorides is attached to **two** electronegative atoms (oxygen and chlorine), and so is polarised δ^+ to a greater extent than the carbon atom in chloroalkanes. This allows it to attract nucleophiles more strongly.

28.4 Polyesters

Formation of polyesters

When a diol is esterified with a diacid (for example, by heating with hydrogen chloride gas), or reacted with a diacyl chloride, a polyester is produced. Figure 28.4 shows the general reaction for esterification with a diacyl chloride. Here the yellow rectangles represent the rest of the diol molecule and the green circles represent the benzene ring in the diacyl chloride.

▼ **Figure 28.4** Formation of a polyester from a diacyl chloride



Notice that the direction of the ester bonds

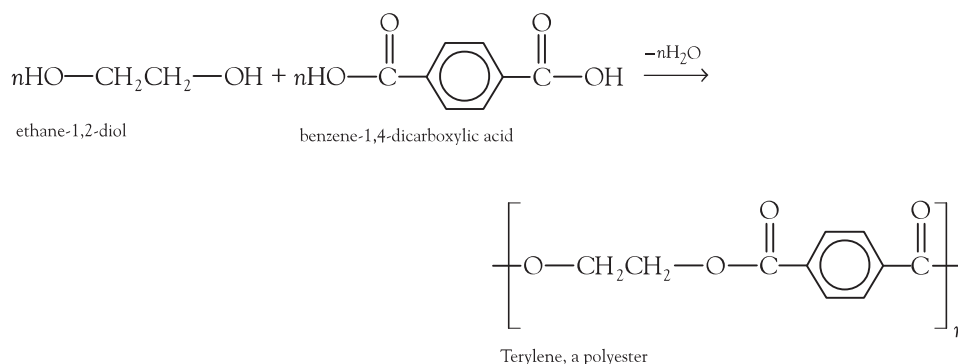


alternates along the chain.

Polyesters are examples of a class of polymers known as **condensation polymers**.

When monomers join together to form a **condensation polymer**, a small molecule such as H_2O , HCl or NH_3 is also produced.

The most commonly used polyester is Terylene, produced by polymerising ethane-1,2-diol and benzene 1,4-dicarboxylic acid, as shown below.



The polymer can be drawn into fine fibres and spun into yarn. Jointly woven with cotton or wool, it is a component of many everyday textiles. Material made from Terylene is hard-wearing and strong, but under strongly alkaline or acidic conditions the ester bonds can be hydrolysed, causing the fabric to break up.



▲ **Figure 28.5** Some products made from polyesters

Further polyesters, and other condensation polymers such as nylon and proteins, are discussed in Topic 30.

SUMMARY

- » The acid strength of **carboxylic acids**, RCO_2H , depends on their structures. In particular, electron-withdrawing chlorine atoms on the alkyl chain near to the CO_2H group increase its acid strength.
- » Methanoic and ethanedioic acids can be oxidised further to CO_2 .
- » **Acyl chlorides** are useful intermediates, forming **esters** with alcohols or phenols, and **amides** with amines.
- » Acyl chlorides are much more easily hydrolysed than chloroalkanes.
- » **Polyesters** are formed by the condensation of a diol with either a dicarboxylic acid or a diacyl chloride.

Key reactions you should know

- » Acyl chlorides:

$$\text{RCOCl} + \text{H}_2\text{O} \rightarrow \text{RCO}_2\text{H} + \text{HCl}$$

$$\text{RCOCl} + \text{R}'\text{OH} \rightarrow \text{RCO}_2\text{R}' + \text{HCl}$$

$$\text{RCOCl} + 2\text{R}'\text{NH}_2 \rightarrow \text{RCONHR}' + \text{R}'\text{NH}_3\text{Cl}$$
 (R' = alkyl or aryl)

- » Condensation polymerisation:

$$n\text{HO}-\text{[orange box]}-\text{OH} + n\text{HO}_2\text{C}-\text{[green circle]}-\text{CO}_2\text{H} \rightarrow$$

$$\text{[O-[orange box]-OCO-[green circle]-CO]}_n$$

END OF TOPIC QUESTIONS

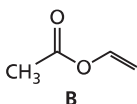
Please see the Data Booklet on the website www.hoddereducation.co.uk/cambridgeextras for any data you may need to answer these questions.

- 1 a Describe and explain how the acidities of $\text{CHCl}_2\text{CO}_2\text{H}$ and $\text{CH}_2\text{ClCO}_2\text{H}$ compare to each other, and to the acidity of ethanoic acid. [3]
- b For each of the following pairs of compounds, suggest one chemical test (reagents and conditions) that would distinguish between them. State the observations you would make with each compound, writing 'none' if appropriate. [7]

First compound	Second compound	Test (reagents and conditions)	Observation with first compound	Observation with second compound
$\text{CH}_3\text{CH}_2\text{COCl}$	$\text{CH}_3\text{COCH}_2\text{Cl}$			
$\text{CH}_3\text{CH}_2\text{CHO}$	CH_3COCH_3			

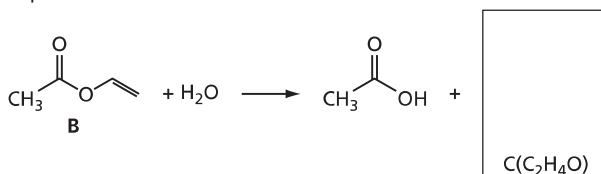
[Cambridge International AS & A Level Chemistry 9701, Paper 42 Q5a&b June 2012]

- 2 a Polyvinyl acetate, PVA, is a useful adhesive for gluing together articles made from wood, paper or cardboard. The monomer of PVA is ethenyl ethanoate, **B**.



PVA is formed from **B** by the process of addition polymerisation.

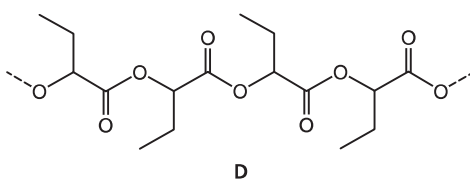
- i Draw a section of the PVA molecule containing at least two monomer molecules, and identify clearly the repeat unit.
- ii The ester **B** can be hydrolysed in the usual way, according to the following equation.



Use this information to suggest a possible structure for **C** and draw it.

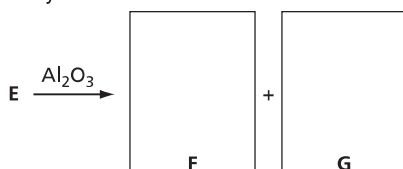
- iii When substance **C** is extracted from the product mixture, it is found that it does not decolourise $\text{Br}_2(\text{aq})$, but it does form a pale yellow precipitate with alkaline aqueous iodine. Suggest a structure for **C** that fits this new information.
- iv Suggest a confirmatory test for the functional group in the structure you have drawn in **iii**. Your answer should include the reagent you would use and the observation you would make. [6]

- b The following diagram represents a section of another polymer



- i Draw brackets, [], around the atoms that make up the repeat unit of this polymer.
- ii Name the functional group in polymer **D**.
- iii Suggest and draw the structure of the monomer, **E**, that could form this polymer.

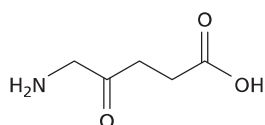
- iv What type of polymerisation is involved in making polymer **D** from its monomer?
- v What is the relationship between the repeat unit of polymer **D** and the repeat unit of PVA?
- c Monomer **E** exists as two stereoisomers. Heating either isomer with Al_2O_3 gives a mixture of two unsaturated carboxylic acids **F** and **G**, which are stereoisomers of each other
- i Name the type of stereoisomerism shown by compound **E**.
- ii Suggest structures for **F** and **G**, and name the type of stereoisomerism they show.



[4]

[Cambridge International AS & A Level Chemistry 9701, Paper 42 Q4 June 2011]

- 3 a Organohalogen compounds can undergo hydrolysis.
 $\text{R-Cl} + \text{H}_2\text{O} \rightarrow \text{R-OH} + \text{HCl}$
 State the relative rates of hydrolysis of the following compounds.
 $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$ $\text{CH}_3\text{CH}_2\text{COCl}$ $\text{C}_6\text{H}_5\text{Cl}$
 Explain your answer.
- b Aminolaevulinic acid is involved in the synthesis of haemoglobin and chlorophyll.

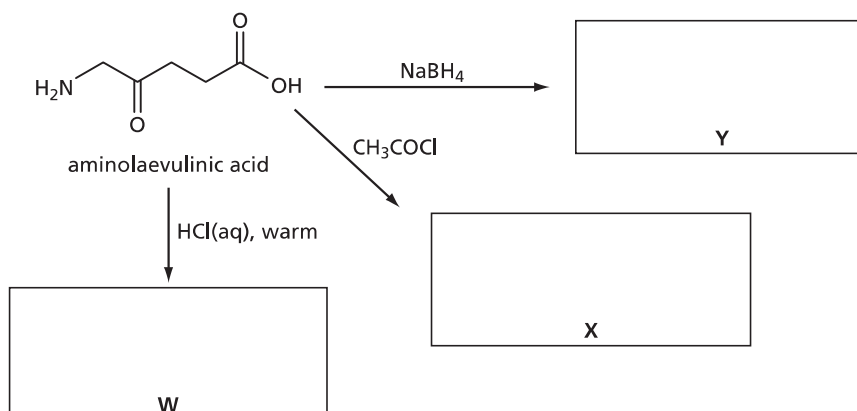


aminolaevulinic acid

Name the **three** functional groups in aminolaevulinic acid.

- c Aminolaevulinic acid reacts readily with bromoethane.
- i Show the mechanism of the **first step** of this reaction on the diagram. Include all necessary curly arrows, lone pairs and relevant dipoles.
-

- ii Name the mechanism in **c i**.
- iii Identify the non-organic product formed in this reaction.
- d Three reactions of aminolaevulinic acid are shown. Draw the structures of the products **W**, **X** and **Y** in the boxes below.

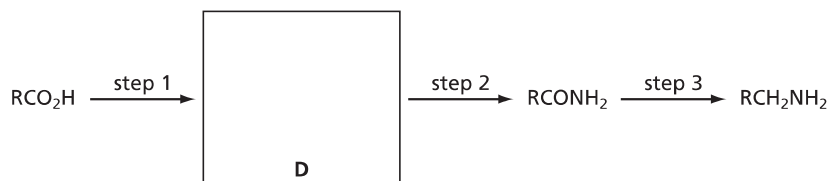


- e Aminolaevulinic acid can undergo polymerisation.

Draw the structure of the polymer showing **two** repeat units. The linkages between the monomer units should be shown fully displayed. [2]

[Cambridge International AS & A Level Chemistry 9701, Paper 41 Q5 October/November 2014]

- 4 a Carboxylic acids can be converted into primary amines by the following sequence of reactions.



- i Suggest the identity of intermediate **D** and write its structure in the box above. [1]
 ii Suggest the reagents for steps 1, 2 and 3. [2]
 b Four compounds, **E**, **F**, **G** and **H**, are isomers of each other. Each compound contains an aromatic ring and **two** functional groups from the following list.
 • alcohol
 • amide
 • amine
 • carboxylic acid
 • ester
 • phenol
 i Which of these functional groups react readily with cold HCl (aq)? [1]
 ii Which of these functional groups react readily with cold NaOH(aq)? [1]

The molecular formula of the four isomers, **E**, **F**, **G** and **H**, is $\text{C}_8\text{H}_9\text{NO}_2$. All four compounds are insoluble in water. Table 1 shows their solubilities in acid or alkali.

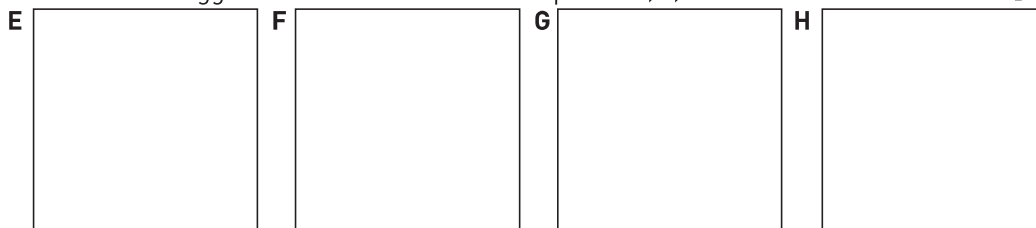
compound	solubility in HCl (aq)	solubility in NaOH(aq)
E	insoluble	insoluble
F	soluble	soluble
G	soluble	insoluble
H	insoluble	soluble

Table 1

- iii Use this information to suggest the **two** functional groups, taken from the list above, that each compound contains. [4]

compound	first functional group	second functional group
E		
F		
G		
H		

- iv Suggest a structure for each compound **E**, **F**, **G** and **H**. [4]



[Cambridge International AS & A Level Chemistry 9701, Paper 42 Q6 May/June 2015]